

Program: Maven Ansible Webapp

Step 1: Create Project

mvn archetype:generate -DgroupId=com.example -DartifactId=AnsibleApp -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false

```
harith@harith-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=AnsibleApp -DarchetypeArtifactId-
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 10 MB/s)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/harith/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: AnsibleApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/harith/devops/AnsibleApp
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 4.468 s
[INFO] Finished at: 2026-05-11T05:51:24+05:30
[INFO]
harith@harith-VirtualBox:~/devops$
```

Step 2: Edit Pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-
v4_0_0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example</groupId>
<artifactId>MavenAnsibleWebApp1</artifactId>
<packaging>war</packaging>
<version>1.0-SNAPSHOT</version>
<name>MavenAnsibleWebApp1</name>
<url>http://maven.apache.org</url>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
<!-- Servlet API for Java EE -->
```

```

<dependency>
<groupId>javax.servlet</groupId>
<artifactId>javax.servlet-api</artifactId>
<version>4.0.1</version>
<scope>provided</scope>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-war-plugin</artifactId>
<version>3.4.0</version>
<!-- Update to latest stable version -->
</plugin>
</plugins>
<finalName>MavenAnsibleWebApp1</finalName>
<!-- create a web page with the name MavenAnsibleWebApp, if we want some other
name we
can chage it -->
</build>
</project>

```

Step 3: Create Ansible Configuration Files

In this step, we create a new folder named ansible inside the project directory using mkdir ansible.

Then, we move into the folder using cd ansible and create the required Ansible files using gedit

hosts.ini and gedit playbook.yml. These files are used to define target hosts and deployment tasks for automation.

```

harith@harith-VirtualBox:~/devops$ cd AnsibleApp/
harith@harith-VirtualBox:~/devops/AnsibleApp$ ls
pom.xml  src
harith@harith-VirtualBox:~/devops/AnsibleApp$ mkdir ansible
harith@harith-VirtualBox:~/devops/AnsibleApp$ gedit hosts.in
harith@harith-VirtualBox:~/devops/AnsibleApp$ ge
geckodriver  genisoimage  geqn          getconf       getent        getkeycodes  getopts      gettext      getty
gedit        genl          getcap       geteltorito  getfacl      getopt       getpcaps     gettext.sh   getweb
harith@harith-VirtualBox:~/devops/AnsibleApp$ gedit playbook.yml
harith@harith-VirtualBox:~/devops/AnsibleApp$ ls
ansible  hosts.in  playbook.yml  pom.xml  src
harith@harith-VirtualBox:~/devops/AnsibleApp$ █

```

gedit host.ini

This command is written inside hosts.ini file to define the Ansible inventory. It creates a group named web and adds localhost as the target system, using local connection and Python 3 interpreter for executing Ansible tasks.

```
localhost ansible_connection=local ansible_python_interpreter=/usr/bin/python3
```

gedit playbook.yml

This code is written inside playbook.yml file to automate deployment using Ansible. It targets the web host group, copies the generated WAR file from the Jenkins workspace to the Tomcat webapps folder, and then restarts the Tomcat service so the updated application runs successfully.

```
- name: Deploy to Tomcat
```

```
hosts: web
```

```
become: yes
```

```
tasks:
```

```
- name: Copy WAR file to Tomcat
```

```
copy:
```

```
src: "/var/snap/jenkins/5013/workspace/MavenAnsibleWebApp1-CICD/target/MavenAnsibleWebApp1.war"
```

```
dest: "/opt/tomcat/webapps/MavenAnsibleWebApp1.war"
```

```
remote_src: no
```

```
- name: Restart Tomcat
```

```
systemd:
```

```
name: tomcat
```

```
state: restarted
```

Step 4: Build the project

In this step, we run the command mvn clean install to clean previous files and build the project. Maven compiles the application, packages it as a .war file, and stores it in the target folder.

```

harith@harith-VirtualBox:~/devops$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:AnsibleApp >-----
[INFO] Building AnsibleApp 1.0-SNAPSHOT
[INFO] -----[ war ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ AnsibleApp ---
[INFO] Deleting /home/harith/devops/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ AnsibleApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/harith/devops/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ AnsibleApp ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ AnsibleApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/harith/devops/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ AnsibleApp ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ AnsibleApp ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ AnsibleApp ---
[INFO] Packaging webapp
[INFO] Assembling webapp [AnsibleApp] in [/home/harith/devops/target/AnsibleApp]
[INFO] Processing war project
[INFO] Building war: /home/harith/devops/target/AnsibleApp.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ AnsibleApp ---
[INFO] Installing /home/harith/devops/target/AnsibleApp.war to /home/harith/.m2/repository/com/example/AnsibleApp/1.0-SNAPSHOT/
[INFO] Installing /home/harith/devops/pom.xml to /home/harith/.m2/repository/com/example/AnsibleApp/1.0-SNAPSHOT/AnsibleApp-1.0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.366 s
[INFO] Finished at: 2026-05-11T06:15:44+05:30
[INFO]

```

Step 5: Jenkins Pipeline

New Item

Enter an item name

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.



Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.



Organization Folder

Creates a set of multibranch project subfolders by scanning for repositories.

OK

Step 6: Select

pipeline Script form SCM

Configure

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/SaiHarithK/ans3.git

Credentials ?

GitHub Credentials

+ Add

Advanced

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

Save

Apply

Step 7: Build Now

Jenkins / MavenAnsibleWebApp1

Status

</> Changes

▶ Build Now

🗑 Delete Pipeline

⚙ Configure

✎ Rename

🔍 Full Stage View

📁 Stages

🔗 Pipeline Syntax

Builds >

🔍 Filter

Today

✅ #4 7:10 am

❌ #3 7:08 am

15 April 2026

❌ #2 10:02 am

❌ #1 9:57 am

❌ MavenAnsibleWebApp1

Stage View

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Archive	Deploy
Average stage times: (full run time: ~30s)	1s	201ms	1s	8s	524ms	10s
#4 May 11 07:10 1 commit	1s	155ms	1s	6s	499ms	16s
#3 May 11 07:08 1 commit	2s	347ms	1s	13s	654ms	7s failed
#2 Apr 15 10:02 No Changes	1s	101ms	1s	5s	359ms	7s failed
#1 Apr 15 09:57 No Changes	1s	202ms	1s	7s	586ms	9s failed

Permalinks

- Last build (#2), 25 days ago
- Last failed build (#2), 25 days ago
- Last unsuccessful build (#2), 25 days ago
- Last completed build (#2), 25 days ago